

QUAID-I-AZAM UNIVERSITY ISLAMABAD
ASSIGNMENT #3



SUBMITTED BY
AWAIS ALI: 04072212045

SUBMITTED TO:
DR. Imran

DEPARTMENT OF COMPUTER SCIENCE
QUAID-I-AZAM UNIVERSITY, ISLAMABAD

TASK# 1

1. **Downloads the latest Minikube binary for Linux.**
`curl -LO https://storage.googleapis.com/minikube/releases/latest/minikube-linux-amd64`
2. **Installs the Minikube executable into system path.**
`minikube start --driver=docker`
3. **Starts a single-node Kubernetes cluster using Docker as the driver.**
`minikube start --driver=docker`
4. **Checks whether Minikube, cluster, and Docker driver are running.**
`minikube status`
5. **Create deployment.yaml file.**
6. **Create deployment using command.**
`kubectl apply -f deployment.yaml`
7. **Shows status of all deployments.**
`kubectl get deployments`
8. **Lists pod details including IP, node, and status.**
`kubectl get pods -o wide`
9. **Displays available nodes in the cluster.**
`kubectl get nodes`
10. **Shows detailed info on node performance and runtime.**
`kubectl describe node <node-name>`
11. **Opens the Kubernetes Dashboard GUI in the browser for visual cluster management.**
`minikube dashboard`

```
awsais@awsais-VirtualBox:~/Documents/devops/assignment3/task1$ curl -LO https://storage.googleapis.com/minikube/releases/latest/minikube-linux-amd64
awsais@awsais-VirtualBox:~/Documents/devops/assignment3/task1$ sudo install minikube-linux-amd64 /usr/local/bin/minikube
awsais@awsais-VirtualBox:~/Documents/devops/assignment3/task1$ minikube version
minikube version: v1.37.0
commit: 65318f4cfff9c12cc87ec9eb8f4cdd57b25047f3
awsais@awsais-VirtualBox:~/Documents/devops/assignment3/task1$
```

```
awsais@awsais-VirtualBox: ~/Documents/devops/assignment3/task1
minikube v1.37.0 on Linuxmint 22.2 (vbox/amd64)
  MINIKUBE_HOME=/home/awsais/.minikube
Using the docker driver based on existing profile
Starting "minikube" primary control-plane node in "minikube" cluster
Pulling base image v0.0.48 ...
docker "minikube" container is missing, will recreate.
Creating docker container (CPUs=2, Memory=3072MB) ...
Preparing Kubernetes v1.34.0 on Docker 28.4.0 ...
Configuring bridge CNI (Container Networking Interface) ...
Verifying Kubernetes components...
  Using image gcr.io/k8s-minikube/storage-provisioner:v5
  Using image docker.io/kubernetes/metrics-scraper:v1.0.8
  Using image docker.io/kubernetes/dashboard:v2.7.0
Some dashboard features require the metrics-server addon. To enable all features please run:
  minikube addons enable metrics-server

Enabled addons: storage-provisioner, default-storageclass, dashboard
Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
awsais@awsais-VirtualBox:~/Documents/devops/assignment3/task1$ minikube status
minikube
type: Control Plane
host: Running
kubelet: Running
apiserver: Running
kubeconfig: Configured

awsais@awsais-VirtualBox:~/Documents/devops/assignment3/task1$ kubectl get nodes
NAME      STATUS   ROLES    AGE   VERSION
minikube  Ready    control-plane  6m3s   v1.34.0
awsais@awsais-VirtualBox:~/Documents/devops/assignment3/task1$ touch deployment.yml
awsais@awsais-VirtualBox:~/Documents/devops/assignment3/task1$ nano deployment.yml
awsais@awsais-VirtualBox:~/Documents/devops/assignment3/task1$ kubectl apply -f deployment.yml
deployment.apps/nginx-deployment created
awsais@awsais-VirtualBox:~/Documents/devops/assignment3/task1$ kubectl get deployments
NAME          READY   UP-TO-DATE   AVAILABLE   AGE
nginx-deployment  0/1     1             0           38s
awsais@awsais-VirtualBox:~/Documents/devops/assignment3/task1$ kubectl get pods -o wide
NAME          READY   STATUS    RESTARTS   AGE   IP          NODE      NOMINATED NODE   READINESS GATES
nginx-deployment-6f9664446b-fn7cl  0/1     ContainerCreating  0      59s   <none>     minikube   <none>           <none>
```

```
awsais@awsais-VirtualBox:~/Documents/devops/assignment3/task1$ kubectl describe pod nginx-deployment-6f9664446b-fn7cl
Name:         nginx-deployment-6f9664446b-fn7cl
Namespace:    default
Priority:      0
Service Account: default
Node:         minikube/192.168.49.2
Start Time:   Sat, 22 Nov 2025 23:52:33 +0500
Labels:       app=nginx
              pod-template-hash=6f9664446b
Annotations:  <none>
Status:       Running
IP:           10.244.0.6
IPs:          10.244.0.6
Controlled By: ReplicaSet/nginx-deployment-6f9664446b
Containers:
  nginx:
    Container ID:  docker://862451888c303a155297e3da5e8ba9b1969b63230f5c8186a9aad2013de58f
    Image:         nginx:latest
    Image ID:      docker-pullable://nginx@sha256:553f64aecdc31b5bf944521731cd70e35da4faed96b2b7548a3d8e2598c52a42
    Port:         80/TCP
    Host Port:     0/TCP
    State:         Running
      Started:     Sat, 22 Nov 2025 23:53:56 +0500
    Ready:         True
    Restart Count: 0
    Environment:   <none>
    Mounts:
      /var/run/secrets/kubernetes.io/serviceaccount from kube-api-access-bmx72 (ro)
Conditions:
  Type              Status
  PodReadyToStartContainers  True
  Initialized         True
  Ready              True
  ContainersReady     True
  PodScheduled        True
Volumes:
  kube-api-access-bmx72:
    Type:              Projected (a volume that contains injected data from multiple sources)
    TokenExpirationSeconds: 3607
    ConfigMapName:      kube-root-ca.crt
    Optional:           false
```

```
awsais@awsais-VirtualBox:~/Documents/devops/assignment3/task1$ kubectl logs nginx-deployment-6f9664446b-fn7cl
/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
/docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
/docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
/docker-entrypoint.sh: Configuration complete; ready for start up
2025/11/22 18:53:57 [notice] 1#1: using the "epoll" event method
2025/11/22 18:53:57 [notice] 1#1: nginx/1.25.3
2025/11/22 18:53:57 [notice] 1#1: built by gcc 14.2.0 (Debian 14.2.0-19)
2025/11/22 18:53:57 [notice] 1#1: OS: Linux 6.14.0-29-generic
2025/11/22 18:53:57 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1048576:1048576
2025/11/22 18:53:57 [notice] 1#1: start worker processes
2025/11/22 18:53:57 [notice] 1#1: start worker process 29
2025/11/22 18:53:57 [notice] 1#1: start worker process 30
2025/11/22 18:53:57 [notice] 1#1: start worker process 31
2025/11/22 18:53:57 [notice] 1#1: start worker process 32
2025/11/22 18:53:57 [notice] 1#1: start worker process 33
2025/11/22 18:53:57 [notice] 1#1: start worker process 34
2025/11/22 18:53:57 [notice] 1#1: start worker process 35
2025/11/22 18:53:57 [notice] 1#1: start worker process 36
awsais@awsais-VirtualBox:~/Documents/devops/assignment3/task1$ kubectl
```

Assignment plan overview
chatgpt.com

ChatGPT
• Inspects & describes deployment, ingress, deployment, ...
• Terminal showing pod created successfully

Enable Kubernetes Dashboard (GUI)
Run:
kubectl enable-dashboard
kubectl dashboard

Workloads

Cron Jobs

Daemon Sets

Deployments

Jobs

Pods

Replica Sets

Replication Controllers

Stateful Sets

Service

Ingresses

Ingress Classes

Services

Config and Storage

Config Maps

Persistent Volume Claims

workload Status

Running: 1

Running: 1

Running: 1

Deployments

Pods

Replica Sets

Deployments

Name	Images	Labels	Pods	Created ↑
nginx-deployment	nginx:latest	-	1 / 1	20 minutes ago

→ ↺ ↻

http://127.0.0.1:34531/api/v1/namespaces/kubernetes-dashboard/services/http:kubernetes-dashboard:/proxy/ 90%

kubernetes

default

Search

+

🔔

Workloads

Workloads

Cron Jobs

Daemon Sets

Deployments

Jobs

Pods

Replica Sets

Replication Controllers

Stateful Sets

Service

Ingresses

Ingress Classes

Services

Config and Storage

Config Maps

Persistent Volume Claims

Deployments

Name	Images	Labels	Pods	Created ↑
nginx-deployment	nginx:latest	-	1 / 1	20 minutes ago

Pods

Name	Images	Labels	Node	Status	Restarts	CPU Usage (cores)	Memory Usage (bytes)	Created ↑
nginx-deployment-6f966444b-fn7cl	nginx:latest	app: nginx pod-template-hash: 6f9664446b	minikube	Running	0	-	-	20 minutes ago

Replica Sets

Name	Images	Labels	Pods	Created ↑
nginx-deployment-6f966444b	nginx:latest	app: nginx pod-template-hash: 6f9664446b	1 / 1	20 minutes ago

TASK#2

1. **Downloads the Kompose binary for Linux.**
`curl -L https://github.com/kubernetes/kompose/releases/latest/download/kompose-linux-amd64 -o kompose`
2. **Makes the Kompose binary executable.**
`chmod +x kompose`
3. **Builds a Docker image for the web application.**
`docker build -t awais3823/webapp:latest .`
4. **Logs in to Docker Hub.**
`docker login`
5. **Pushes the built image to Docker Hub.**
`docker push awais3823/webapp:latest`
6. **Converts docker-compose.yml into Kubernetes YAML files using Kompose.**
`kompose convert`
7. **Starts Minikube with Docker driver.**
`minikube start --driver=docker`
8. **Deploys the web application Deployment to Kubernetes.**
`kubectl apply -f web-deployment.yaml`
9. **Deploys the Redis Deployment to Kubernetes.**
`kubectl apply -f redis-deployment.yaml`
10. **Deploys the web service.**
`kubectl apply -f web-service.yaml`
11. **Deploys the Redis service.**
`kubectl apply -f redis-service.yaml`
12. **Lists all pods running in the cluster.**
`kubectl get pods`
13. **Lists all Kubernetes services.**
`kubectl get svc`
14. **Opens the running web application service in the browser.**
`minikube service web`

```
awais@awais-VirtualBox: ~/Documents/devops/assignment3/task2
awais@awais-VirtualBox:~/Documents/devops/assignment3/task2$ docker build -t awais3823/python-web .
[+] Building 38.0s (10/10) FINISHED
=> [internal] load build definition from Dockerfile                                docker:default 0.3s
=> => transferring dockerfile: 368B                                              0.1s
=> [internal] load metadata for docker.io/library/python:3.10-slim              7.8s
=> [auth] library/python:pull token for registry-1.docker.io                   0.0s
=> [internal] load .dockerignore                                                 0.3s
=> => transferring context: 2B                                                  0.1s
=> [1/4] FROM docker.io/library/python:3.10-slim@sha256:c299e10e0070171113f9a1f109dd05e7e634fa94589b056e0e87bb22b2b382a2 0.0s
=> [internal] load build context                                                0.3s
=> => transferring context: 902B                                              0.1s
=> CACHED [2/4] WORKDIR /code                                                  0.0s
=> [3/4] COPY . /code                                                         0.5s
=> [4/4] RUN pip install --no-cache-dir flask redis                          24.5s
=> exporting to image                                                         2.8s
=> => exporting layers                                                         2.3s
=> => writing image sha256:60c3c1c1a16f70b23686dc97eb0730e50bd16412ffded01ed0af324f3fa72197 0.1s
=> => naming to docker.io/awais3823/python-web                                0.1s
```

```
=> naming to docker.io/awais3823/python-web
awais@awais-VirtualBox:~/Documents/devops/assignment3/task2$ docker push awais3823/python-web
Using default tag: latest
The push refers to repository [docker.io/awais3823/python-web]
286704c89954: Pushed
204a364227a6: Pushed
492fbc8d8763f: Layer already exists
548bale7e829: Layer already exists
a8045a14b5f4: Layer already exists
cb4ecf39d967: Layer already exists
70a290c5e58b: Layer already exists
latest: digest: sha256:d20e9a51f67561cc4e6f1e02c99c46b16239bf50a0b942d511ff4cbb6b7de1e2 size: 1784
awais@awais-VirtualBox:~/Documents/devops/assignment3/task2$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
nginx-deployment-6f9664446b-fn7cl  1/1     Running   6           24h
redis-694874dd5f-7pgfc             1/1     Running   0           57m
web-554f574778-vz556               0/1     CrashLoopBackOff   4 (4m57s ago)  9m3s
awais@awais-VirtualBox:~/Documents/devops/assignment3/task2$ kubectl set image deployment/web web=awais3823/python-web
awais@awais-VirtualBox:~/Documents/devops/assignment3/task2$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
nginx-deployment-6f9664446b-fn7cl  1/1     Running   6           24h
redis-694874dd5f-7pgfc             1/1     Running   0           60m
web-554f574778-vz556               1/1     Running   5           12m
awais@awais-VirtualBox:~/Documents/devops/assignment3/task2$ kubectl get svc
NAME      TYPE        CLUSTER-IP   EXTERNAL-IP   PORT(S)   AGE
kubernetes  ClusterIP   10.96.0.1    <none>        443/TCP   24h
```

```
=> CACHED [2/4] WORKDIR /code
=> [3/4] COPY /code
=> [4/4] RUN pip install --no-cache-dir flask redis
=> exporting to image
=> exporting layers
=> writing image sha256:60c3c1cba16f70b23686dc97eb0730e50bd16412ffed01ed0af324f3fa71297
=> naming to docker.io/awais3823/python-web
awais@awais-VirtualBox:~/Documents/devops/assignment3/task2$ docker push awais3823/python-web
Using default tag: latest
The push refers to repository [docker.io/awais3823/python-web]
286704c89954: Pushed
204a364227a6: Pushed
492fbc8d8763f: Layer already exists
548bale7e829: Layer already exists
a8045a14b5f4: Layer already exists
cb4ecf39d967: Layer already exists
70a290c5e58b: Layer already exists
latest: digest: sha256:d20e9a51f67561cc4e6f1e02c99c46b16239bf50a0b942d511ff4cbb6b7de1e2 size: 1784
awais@awais-VirtualBox:~/Documents/devops/assignment3/task2$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
nginx-deployment-6f9664446b-fn7cl  1/1     Running   6           24h
redis-694874dd5f-7pgfc             1/1     Running   0           57m
web-554f574778-vz556               0/1     CrashLoopBackOff   4 (4m57s ago)  9m3s
awais@awais-VirtualBox:~/Documents/devops/assignment3/task2$ kubectl set image deployment/web web=awais3823/python-web
awais@awais-VirtualBox:~/Documents/devops/assignment3/task2$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
nginx-deployment-6f9664446b-fn7cl  1/1     Running   6           24h
redis-694874dd5f-7pgfc             1/1     Running   0           60m
web-554f574778-vz556               1/1     Running   5           12m
awais@awais-VirtualBox:~/Documents/devops/assignment3/task2$ kubectl get svc
NAME      TYPE        CLUSTER-IP   EXTERNAL-IP   PORT(S)   AGE
kubernetes  ClusterIP   10.96.0.1    <none>        443/TCP   24h
redis      ClusterIP   10.99.192.126 <none>        6379/TCP   7h30m
web        NodePort    10.96.34.29   <none>        5000:30080/TCP 7h30m
awais@awais-VirtualBox:~/Documents/devops/assignment3/task2$ minikube service web

```

NAMESPACE	NAME	TARGET PORT	URL
default	web	5000/5000	http://192.168.49.2:30080

```
Opening service default/web in default browser...
awais@awais-VirtualBox:~/Documents/devops/assignment3/task2$
```



TASK#3

1. **Deletes the old Minikube cluster to start fresh.**
`minikube delete`
2. **Starts a new Minikube cluster with 2 nodes.**
`minikube start --nodes 2`
3. **Waits until both nodes are fully ready.**
`kubectrl wait --for=condition=Ready node --all --timeout=180s`
4. **Displays all nodes to confirm readiness.**
`kubectrl get nodes`
5. **Applies the original deployment configuration.**
`kubectrl apply -f qu3/hello-deployment.yaml`
6. **Shows the running pods and which node they are on.**
`kubectrl get pods -o wide`
7. **Deletes the previously deployed application.**
`kubectrl delete deployment hello`
8. **Labels the second node with disktype=ssd for scheduling.**
`kubectrl label nodes <node-name> disktype=ssd --overwrite`
(In script: node name is fetched automatically.)
9. **Displays all node labels to confirm labeling.**
`kubectrl get nodes --show-labels`
10. **Creates an updated deployment YAML with replicas=2 and nodeSelector.**
(hello-deployment_updated.yaml is generated using a here-document.)
11. **Applies the updated deployment to schedule pods on the labeled node.**
`kubectrl apply -f hello-deployment_updated.yaml`
12. **Checks the pods and shows which node they are running on.**
`kubectrl get pods -o wide`


```
aws@aws-VirtualBox: ~/Documents/devops/assignment3/task3
File Edit View Search Terminal Help
minikube-m02 Ready <none> 26s v1.34.0
aws@aws-VirtualBox:~/Documents/devops/assignment3/task3$ kubectl delete deployment hello
Error from server (NotFound): deployments.apps "hello" not found
aws@aws-VirtualBox:~/Documents/devops/assignment3/task3$ kubectl get pods
No resources found in default namespace.
aws@aws-VirtualBox:~/Documents/devops/assignment3/task3$ kubectl get pods
No resources found in default namespace.
aws@aws-VirtualBox:~/Documents/devops/assignment3/task3$ kubectl get nodes
NAME STATUS ROLES AGE VERSION
minikube Ready control-plane 24m v1.34.0
minikube-m02 Ready <none> 19m v1.34.0
aws@aws-VirtualBox:~/Documents/devops/assignment3/task3$ kubectl label nodes minikube-m02 disktype=ssd
node/minikube-m02 labeled
aws@aws-VirtualBox:~/Documents/devops/assignment3/task3$ kubectl get nodes --show-labels
NAME STATUS ROLES AGE VERSION LABELS
minikube Ready control-plane 26m v1.34.0 beta.kubernetes.io/arch=amd64,beta.kubernetes.io/os=linux,kubernetes.io/arch=amd64,kubernetes.io/hostname=minikube,kubernetes.io/os=linux,minikube.k8s.io/commit=65318f4cfff9c12cc87ec9eb8f4cdd57b25047f3,minikube.k8s.io/name=minikube,minikube.k8s.io/primary=true,minikube.k8s.io/updated_at=2025_11_24T01_01_38_0700,minikube.k8s.io/version=v1.37.0,node-role.kubernetes.io/control-plane=,node.kubernetes.io/exclude-from-external-load-balancers=
minikube-m02 Ready <none> 21m v1.34.0 beta.kubernetes.io/arch=amd64,beta.kubernetes.io/os=linux,disktype=ssd,kubernetes.io/arch=amd64,kubernetes.io/hostname=minikube-m02,kubernetes.io/os=linux,minikube.k8s.io/commit=65318f4cfff9c12cc87ec9eb8f4cdd57b25047f3,minikube.k8s.io/name=minikube,minikube.k8s.io/primary=false,minikube.k8s.io/updated_at=2025_11_24T01_05_30_0700,minikube.k8s.io/version=v1.37.0
aws@aws-VirtualBox:~/Documents/devops/assignment3/task3$ kubectl label nodes minikube-m02 disktype=
node/minikube-m02 unlabeled
aws@aws-VirtualBox:~/Documents/devops/assignment3/task3$ kubectl label nodes minikube-m02 disktype=
label "disktype" not found.
node/minikube-m02 not labeled
aws@aws-VirtualBox:~/Documents/devops/assignment3/task3$ kubectl apply -f hello-deployment.yaml
error: the path "hello-deployment.yaml" does not exist
aws@aws-VirtualBox:~/Documents/devops/assignment3/task3$ kubectl apply -f hello-deployment.yaml
deployment.apps/hello created
aws@aws-VirtualBox:~/Documents/devops/assignment3/task3$ kubectl get pods -o wide
NAME READY STATUS RESTARTS AGE IP NODE NOMINATED NODE READINESS GATES
hello-bd796f95b-78txn 0/1 ImagePullBackOff 0 64s 10.244.1.2 minikube-m02 <none> <none>
hello-bd796f95b-wmlwb 0/1 ContainerCreating 0 63s <none> minikube <none> <none>
aws@aws-VirtualBox:~/Documents/devops/assignment3/task3$ kubectl delete deployment hello
deployment.apps "hello" deleted from default namespace
aws@aws-VirtualBox:~/Documents/devops/assignment3/task3$ kubectl get pods
NAME READY STATUS RESTARTS AGE
hello-bd796f95b-78txn 0/1 Terminating 0 3m38s
```

```
aws@aws-VirtualBox: ~/Documents/devops/assignment3/task3
File Edit View Search Terminal Help
minikube-m02 Ready <none> 26s v1.34.0
aws@aws-VirtualBox:~/Documents/devops/assignment3/task3$ kubectl delete deployment hello
Error from server (NotFound): deployments.apps "hello" not found
aws@aws-VirtualBox:~/Documents/devops/assignment3/task3$ kubectl get pods
No resources found in default namespace.
aws@aws-VirtualBox:~/Documents/devops/assignment3/task3$ kubectl get pods
No resources found in default namespace.
aws@aws-VirtualBox:~/Documents/devops/assignment3/task3$ kubectl get nodes
NAME STATUS ROLES AGE VERSION
minikube Ready control-plane 24m v1.34.0
minikube-m02 Ready <none> 19m v1.34.0
aws@aws-VirtualBox:~/Documents/devops/assignment3/task3$ kubectl label nodes minikube-m02 disktype=ssd
node/minikube-m02 labeled
aws@aws-VirtualBox:~/Documents/devops/assignment3/task3$ kubectl get nodes --show-labels
NAME STATUS ROLES AGE VERSION LABELS
minikube Ready control-plane 26m v1.34.0 beta.kubernetes.io/arch=amd64,beta.kubernetes.io/os=linux,kubernetes.io/arch=amd64,kubernetes.io/hostname=minikube,kubernetes.io/os=linux,minikube.k8s.io/commit=65318f4cfff9c12cc87ec9eb8f4cdd57b25047f3,minikube.k8s.io/name=minikube,minikube.k8s.io/primary=true,minikube.k8s.io/updated_at=2025_11_24T01_01_38_0700,minikube.k8s.io/version=v1.37.0,node-role.kubernetes.io/control-plane=,node.kubernetes.io/exclude-from-external-load-balancers=
minikube-m02 Ready <none> 21m v1.34.0 beta.kubernetes.io/arch=amd64,beta.kubernetes.io/os=linux,disktype=ssd,kubernetes.io/arch=amd64,kubernetes.io/hostname=minikube-m02,kubernetes.io/os=linux,minikube.k8s.io/commit=65318f4cfff9c12cc87ec9eb8f4cdd57b25047f3,minikube.k8s.io/name=minikube,minikube.k8s.io/primary=false,minikube.k8s.io/updated_at=2025_11_24T01_05_30_0700,minikube.k8s.io/version=v1.37.0
aws@aws-VirtualBox:~/Documents/devops/assignment3/task3$ kubectl label nodes minikube-m02 disktype=
node/minikube-m02 unlabeled
aws@aws-VirtualBox:~/Documents/devops/assignment3/task3$ kubectl label nodes minikube-m02 disktype=
label "disktype" not found.
node/minikube-m02 not labeled
aws@aws-VirtualBox:~/Documents/devops/assignment3/task3$ kubectl apply -f hello-deployment.yaml
error: the path "hello-deployment.yaml" does not exist
aws@aws-VirtualBox:~/Documents/devops/assignment3/task3$ kubectl apply -f hello-deployment.yaml
deployment.apps/hello created
aws@aws-VirtualBox:~/Documents/devops/assignment3/task3$ kubectl get pods -o wide
NAME READY STATUS RESTARTS AGE IP NODE NOMINATED NODE READINESS GATES
hello-bd796f95b-78txn 0/1 ImagePullBackOff 0 64s 10.244.1.2 minikube-m02 <none> <none>
hello-bd796f95b-wmlwb 0/1 ContainerCreating 0 63s <none> minikube <none> <none>
aws@aws-VirtualBox:~/Documents/devops/assignment3/task3$ kubectl delete deployment hello
deployment.apps "hello" deleted from default namespace
aws@aws-VirtualBox:~/Documents/devops/assignment3/task3$ kubectl get pods
NAME READY STATUS RESTARTS AGE
hello-bd796f95b-78txn 0/1 Terminating 0 3m38s
```



```
awais@awais-VirtualBox: ~/Documents/devops/assignment3/task3
File Edit View Search Terminal Help
pod hello-bd796f95b-wmlwb deleted from default namespace
awais@awais-VirtualBox:~/Documents/devops/assignment3/task3$ kubectl get pods
NAME          READY   STATUS    RESTARTS   AGE
hello-bd796f95b-wmlwb  0/1     Terminating    0          9m55s
awais@awais-VirtualBox:~/Documents/devops/assignment3/task3$ kubectl delete pod --all --grace-period=0 --force
Warning: Immediate deletion does not wait for confirmation that the running resource has been terminated. The resource may continue to run on the cluster indefinitely.
pod "hello-bd796f95b-wmlwb" force deleted from default namespace
awais@awais-VirtualBox:~/Documents/devops/assignment3/task3$ kubectl get pods
No resources found in default namespace.
awais@awais-VirtualBox:~/Documents/devops/assignment3/task3$ kubectl get nodes
NAME          STATUS   ROLES    AGE   VERSION
minikube      Ready    control-plane  60m   v1.34.0
minikube-m02  Ready    <none>      56m   v1.34.0
awais@awais-VirtualBox:~/Documents/devops/assignment3/task3$ kubectl label nodes minikube-m02 disktype=ssd
node/minikube-m02 labeled
awais@awais-VirtualBox:~/Documents/devops/assignment3/task3$ kubectl get nodes --show-labels
NAME          STATUS   ROLES    AGE   VERSION   LABELS
minikube      Ready    control-plane  61m   v1.34.0   beta.kubernetes.io/arch=amd64,beta.kubernetes.io/os=linux,kubernetes.io/arch=amd64,kubernetes.io/hostname=minikube,kubernetes.io/os=linux,minikube.k8s.io/commit=65318f4cfff9c12cc87ec9eb8f4cdd57b25047f3,minikube.k8s.io/name=minikube,minikube.k8s.io/primary=true,minikube.k8s.io/updated_at=2025_11_24T01_01_38_0700,minikube.k8s.io/version=v1.37.0,node-role.kubernetes.io/control-plane=node.kubernetes.io/exclude-from-external-load-balancers=
minikube-m02  Ready    <none>      57m   v1.34.0   beta.kubernetes.io/arch=amd64,beta.kubernetes.io/os=linux,disktype=ssd,kubernetes.io/arch=amd64,kubernetes.io/hostname=minikube-m02,kubernetes.io/os=linux,minikube.k8s.io/commit=65318f4cfff9c12cc87ec9eb8f4cdd57b25047f3,minikube.k8s.io/name=minikube,minikube.k8s.io/primary=false,minikube.k8s.io/updated_at=2025_11_24T01_05_30_0700,minikube.k8s.io/version=v1.37.0
awais@awais-VirtualBox:~/Documents/devops/assignment3/task3$ kubectl apply -f hello-deployment_updated.yaml
deployment.apps/hello created
awais@awais-VirtualBox:~/Documents/devops/assignment3/task3$ kubectl get pods -o wide
NAME          READY   STATUS    RESTARTS   AGE   IP          NODE          NOMINATED NODE   READINESS GATES
hello-6f4fd7f4cf-c88xw  0/1     ImagePullBackOff    0      49s   10.244.1.3   minikube-m02   <none>           <none>
hello-6f4fd7f4cf-gjk68  0/1     ImagePullBackOff    0      48s   10.244.1.4   minikube-m02   <none>           <none>
awais@awais-VirtualBox:~/Documents/devops/assignment3/task3$ kubectl describe deployment hello
Name:
Namespace:
CreationTimestamp:
Labels:
Annotations:
Selector:
Replicas:
StrategyType:
MinReadySeconds:
RollingUpdateStrategy:
2 desired | 2 updated | 2 total | 0 available | 2 unavailable
RollingUpdate
```

```
awais@awais-VirtualBox: ~/Documents/devops/assignment3/task3
File Edit View Search Terminal Help
hello-6f4fd7f4cf-c88xw  0/1     ImagePullBackOff    0      49s   10.244.1.3   minikube-m02   <none>           <none>
hello-6f4fd7f4cf-gjk68  0/1     ImagePullBackOff    0      48s   10.244.1.4   minikube-m02   <none>           <none>
awais@awais-VirtualBox:~/Documents/devops/assignment3/task3$ kubectl describe deployment hello
Name:
Namespace:
CreationTimestamp:
Labels:
Annotations:
Selector:
Replicas:
StrategyType:
MinReadySeconds:
RollingUpdateStrategy:
25% max unavailable, 25% max surge
Pod Template:
Labels: app=hello
Containers:
hello-from:
Image:
Port:
Host Port:
Environment:
Mounts:
Volumes:
Node-Selectors:
Tolerations:
Conditions:
Type
Status
Reason
Available
False
MinimumReplicasUnavailable
Progressing
True
ReplicaSetUpdated
OldReplicaSets:
NewReplicaSet:
Events:
Type
Reason
Age
From
Message
Normal
ScalingReplicaSet
110s
deployment-controller
Scaled up replica set hello-6f4fd7f4cf from 0 to 2
awais@awais-VirtualBox:~/Documents/devops/assignment3/task3$ kubectl get pods -o wide --sort-by=.spec.nodeName
NAME          READY   STATUS    RESTARTS   AGE   IP          NODE          NOMINATED NODE   READINESS GATES
hello-6f4fd7f4cf-c88xw  0/1     ImagePullBackOff    0      2m30s   10.244.1.3   minikube-m02   <none>           <none>
hello-6f4fd7f4cf-gjk68  1/1     Running    0          2m29s   10.244.1.4   minikube-m02   <none>           <none>
awais@awais-VirtualBox:~/Documents/devops/assignment3/task3$
```

TASK#4

1. **Starts Minikube using Docker driver.**
`minikube start --driver=docker`
2. **Applies the Persistent Volume (PV) configuration for MySQL storage.**
`kubectl apply -f mysql-pv.yaml`
3. **Applies the Persistent Volume Claim (PVC) to request storage.**
`kubectl apply -f mysql-pvc.yaml`
4. **Deploys the MySQL server using a Deployment YAML.**
`kubectl apply -f mysql-deployment.yaml`
5. **Exposes the MySQL database internally using a ClusterIP service.**
`kubectl apply -f mysql-service.yaml`
6. **Deploys PhpMyAdmin to manage the MySQL database.**
`kubectl apply -f phpmyadmin-deployment.yaml`
7. **Exposes PhpMyAdmin externally using a NodePort service.**
`kubectl apply -f phpmyadmin-service.yaml`
8. **Lists all running pods to verify deployments.**
`kubectl get pods`
9. **Lists all services including MySQL and PhpMyAdmin.**
`kubectl get svc`
10. **Checks the status of Persistent Volumes.**
`kubectl get pv`
11. **Checks the status of Persistent Volume Claims.**
`kubectl get pvc`
12. **Verifies that the MySQL pod has mounted the PVC successfully.**
`kubectl describe pod <mysql-pod-name>`
13. **Opens PhpMyAdmin in the browser via Minikube.**
`minikube service phpmyadmin`
14. **Enters the MySQL pod for database access.**
`kubectl exec -it <mysql-pod-name> -- bash`
15. **Logs in to MySQL inside the container.**
`mysql -u root -p`
16. **Creates a test database to check persistence.**
`CREATE DATABASE testdb;`

17. Deletes the MySQL pod to test data persistence after restart.

```
kubectl delete pod <mysql-pod-name>
```

18. Verifies that the database still exists in the new pod after restart.

```
kubectl exec -it <new-mysql-pod-name> -- mysql -u root -p
```

```
awais@awais-VirtualBox: ~/Documents/devops/assignment3/task4
minikube v1.37.0 on Linuxmint 22.2 (vbox/amd64)
  MINIKUBE_HOME=/home/awais/.minikube
Using the Docker driver based on existing profile
Starting "minikube" primary control-plane node in "minikube" cluster
Pulling base image v0.0.48 ...
Restarting existing docker container for "minikube" ...
Preparing Kubernetes v1.34.0 on Docker 28.4.0 ...
Verifying Kubernetes components...
Enabled addons:

Starting "minikube-m02" worker node in "minikube" cluster
Pulling base image v0.0.48 ...
Restarting existing docker container for "minikube-m02" ...
Found network options:
  NO_PROXY=192.168.49.2
Preparing Kubernetes v1.34.0 on Docker 28.4.0 ...
  env NO_PROXY=192.168.49.2
Verifying Kubernetes components...
Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
awais@awais-VirtualBox:~/Documents/devops/assignment3/task4$ kubectl apply -f mysql-pv.yaml
kubectl apply -f mysql-pvc.yaml
kubectl apply -f mysql-deployment.yaml
kubectl apply -f mysql-service.yaml

kubectl apply -f phpmyadmin-deployment.yaml
kubectl apply -f phpmyadmin-service.yaml
error: error validating "mysql-pv.yaml": error validating data: failed to download openapi: Get "https://192.168.49.2:8443/openapi/v2?timeout=32s": dial tcp 192.168.49.2:8443: connect: no route to host; if you choose to ignore these errors, turn validation off with --validate=false
error: error validating "mysql-pvc.yaml": error validating data: failed to download openapi: Get "https://192.168.49.2:8443/openapi/v2?timeout=32s": dial tcp 192.168.49.2:8443: connect: no route to host; if you choose to ignore these errors, turn validation off with --validate=false
error: error validating "mysql-deployment.yaml": error validating data: failed to download openapi: Get "https://192.168.49.2:8443/openapi/v2?timeout=32s": dial tcp 192.168.49.2:8443: connect: no route to host; if you choose to ignore these errors, turn validation off with --validate=false
error: error validating "mysql-service.yaml": error validating data: failed to download openapi: Get "https://192.168.49.2:8443/openapi/v2?timeout=32s": dial tcp 192.168.49.2:8443: connect: no route to host; if you choose to ignore these errors, turn validation off with --validate=false
error: error validating "phpmyadmin-deployment.yaml": error validating data: failed to download openapi: Get "https://192.168.49.2:8443/openapi/v2?timeout=32s": dial tcp 192.168.49.2:8443: connect: no route to host; if you choose to ignore
bash: awais@awais-VirtualBox:~/Documents/devops/assignment3/task4$: No such file or directory
persistentvolumeclaim/mysql-pvc created
deployment.apps/mysql created
service/mysql created
deployment.apps/phpmyadmin created
service/phpmyadmin created
```

```
awais@awais-VirtualBox:~/Documents/devops/assignment3/task4$ kubectl apply -f mysql-pv.yaml
kubectl apply -f mysql-pvc.yaml
kubectl apply -f mysql-deployment.yaml
kubectl apply -f mysql-service.yaml

kubectl apply -f phpmyadmin-deployment.yaml
kubectl apply -f phpmyadmin-service.yaml
persistentvolumeclaim/mysql-pv unchanged
persistentvolumeclaim/mysql-pvc unchanged
deployment.apps/mysql unchanged
service/mysql unchanged
deployment.apps/phpmyadmin unchanged
service/phpmyadmin unchanged
awais@awais-VirtualBox:~/Documents/devops/assignment3/task4$ kubectl get pods
NAME                                READY    STATUS    RESTARTS   AGE
hello-6f4fd7f4cf-c88xw              1/1      Running   1 (3h59m ago)    40h
hello-6f4fd7f4cf-gjk68              1/1      Running   1              40h
mysql-6974fc57d-cv6l6               1/1      Running   0              3h57m
phpmyadmin-75bf46b7d5-b6f75         1/1      Running   0              3h57m
awais@awais-VirtualBox:~/Documents/devops/assignment3/task4$ kubectl get svc
NAME      TYPE      CLUSTER-IP      EXTERNAL-IP      PORT(S)      AGE
kubernetes  ClusterIP  10.96.0.1        <none>            443/TCP      41h
mysql      ClusterIP  10.97.216.30     <none>            3306/TCP      3h58m
phpmyadmin NodePort    10.105.217.135   <none>            80:30080/TCP  3h58m
awais@awais-VirtualBox:~/Documents/devops/assignment3/task4$ kubectl get pv
NAME      CAPACITY   ACCESS MODES   RECLAIM POLICY   STATUS   CLAIM                STORAGECLASS   VOLUMEATTRIBUTESCLASS   R
mysql-pv  2Gi        RWO            Retain           Available      default/mysql-pvc    standard        <unset>
pvc-e6afd3eb-0064-4d7a-9197-5120bbec20c5  2Gi        RWO            Delete           Bound         default/mysql-pvc    standard        <unset>
awais@awais-VirtualBox:~/Documents/devops/assignment3/task4$ kubectl get pvc
NAME      STATUS   VOLUME                                     CAPACITY   ACCESS MODES   STORAGECLASS   VOLUMEATTRIBUTESCLASS   AGE
mysql-pvc Bound    pvc-e6afd3eb-0064-4d7a-9197-5120bbec20c5  2Gi        RWO            standard        <unset>              3h59m
awais@awais-VirtualBox:~/Documents/devops/assignment3/task4$ minikube service phpmyadmin

  NAMESPACE  NAME      TARGET PORT  URL
  default    phpmyadmin  80           http://192.168.49.2:30080
Opening service default/phpmyadmin in default browser...
awais@awais-VirtualBox:~/Documents/devops/assignment3/task4$
```

TASK#5

1. **Starts Minikube using Docker driver.**
`minikube start --driver=docker`
2. **Creates a ConfigMap containing application configuration values.**
`kubectl apply -f app-configmap.yaml`
3. **Creates a Secret to store sensitive information securely.**
`kubectl apply -f app-secret.yaml`
4. **Deploys the application and injects ConfigMap + Secret as environment variables.**
`kubectl apply -f app-deployment.yaml`
5. **Exposes the application internally using a ClusterIP service.**
`kubectl apply -f app-service.yaml`
6. **Lists all ConfigMaps to verify creation.**
`kubectl get configmaps`
7. **Lists all Secrets in the namespace.**
`kubectl get secrets`
8. **Shows all application deployments running in the cluster.**
`kubectl get deployments`
9. **Lists all pods to verify the application was created.**
`kubectl get pods`
10. **Describes the pod to confirm injected environment variables.**
`kubectl describe pod <pod-name>`
11. **Enters the running pod shell for inspection.**
`kubectl exec -it <pod-name> -- sh`
12. **Prints all environment variables inside the pod.**
`printenv`
13. **Exits from the pod shell.**
`exit`

```
awsais@awsais-VirtualBox: ~/Documents/devops/assignment3/task5
awsais@awsais-VirtualBox:~/Documents/devops/assignment3/task5$ kubectl apply -f app-configmap.yaml
configmap/app-config created
awsais@awsais-VirtualBox:~/Documents/devops/assignment3/task5$ kubectl get configmap
NAME      DATA      AGE
app-config 2          44s
kube-root-ca.crt 1          42h
awsais@awsais-VirtualBox:~/Documents/devops/assignment3/task5$ kubectl describe configmap app-config
Name:      app-config
Namespace: default
Labels:    <none>
Annotations: <none>

Data
====
APP_ENV:
----
production

APP_NAME:
----
MyK8sApp

BinaryData
====
Events: <none>
awsais@awsais-VirtualBox:~/Documents/devops/assignment3/task5$ kubectl apply -f app-secret.yaml
secret/app-secret created
awsais@awsais-VirtualBox:~/Documents/devops/assignment3/task5$ kubectl get secret
NAME      TYPE      DATA      AGE
app-secret Opaque    2          27s
awsais@awsais-VirtualBox:~/Documents/devops/assignment3/task5$ kubectl describe secret app-secret
Name:      app-secret
Namespace: default
Labels:    <none>
Annotations: <none>

Type: Opaque

Data
====
DR PASSWORD: 8 bytes
```

```
awsais@awsais-VirtualBox: ~/Documents/devops/assignment3/task5
Normal Pulling 11m kubelet Pulling image "nginx"
awsais@awsais-VirtualBox:~/Documents/devops/assignment3/task5$ kubectl get pods
NAME                                READY    STATUS    RESTARTS   AGE
demo-app-5785849cf5-lvfc6          0/1      ContainerCreating    0          14m
hello-6f4fd7f4cf-c88xw             1/1      Running        1 (5h3m ago)    41h
hello-6f4fd7f4cf-gjk68             1/1      Running        1              41h
mysql-6974fc57d-cv6l6              1/1      Running        0             5h
phpmyadmin-75bf46b7d5-b6f75        1/1      Running        0             5h
awsais@awsais-VirtualBox:~/Documents/devops/assignment3/task5$ kubectl get events --sort-by='.lastTimestamp'
LAST SEEN   TYPE      REASON          OBJECT                                MESSAGE
15m         Normal    Scheduled        pod/demo-app-5785849cf5-lvfc6        Successfully assigned default/demo-app-5785849cf5-lvfc6 to minikube-m02
15m         Normal    ScalingReplicaSet deployment/demo-app        Scaled up replica set demo-app-5785849cf5 from 0 to 1
15m         Normal    SuccessfulCreate replicaset/demo-app-5785849cf5    Created pod: demo-app-5785849cf5-lvfc6
14m         Normal    Pulling          pod/demo-app-5785849cf5-lvfc6        Pulling image "nginx"
awsais@awsais-VirtualBox:~/Documents/devops/assignment3/task5$ minikube ssh
Error: unknown command "ssh" for "minikube"

Did you mean this?
  ssh

Run 'minikube --help' for usage.
awsais@awsais-VirtualBox:~/Documents/devops/assignment3/task5$ minikube ssh
docker@minikube:~$ sudo docker pull nginx
Using default tag: latest
latest: Pulling from library/nginx
0e4bc2bd6656: Pull complete
b5feb73171bf: Pull complete
108ab8292820: Pull complete
53d743880af4: Pull complete
77fa2eb06317: Pull complete
192e2451f875: Pull complete
de57a609c9d5: Pull complete
Digest: sha256:553f64aecd31b5bf944521731cd70e35da4faed96b2b7548a3d8e2598c52a42
Status: Downloaded newer image for nginx:latest
docker.io/library/nginx:latest
docker@minikube:~$ ^C
docker@minikube:~$ ^C
docker@minikube:~$ minikube exist ssh
-bash: minikube: command not found
docker@minikube:~$ exit
logout
ssh: Process exited with status 127
awsais@awsais-VirtualBox:~/Documents/devops/assignment3/task5$ kubectl net pods
```

```
awais@awais-VirtualBox: ~/Documents/devops/assignment3/task5
Error: unknown command "shh" for "minikube"
Did you mean this?
    ssh

Run 'minikube --help' for usage.
awais@awais-VirtualBox:~/Documents/devops/assignment3/task5$ minikube ssh
docker@minikube:~$ sudo docker pull nginx
Using default tag: latest
latest: Pulling from library/nginx
0e4bc2bd6656: Pull complete
b5feb73171bf: Pull complete
108ab8292820: Pull complete
53d743880af4: Pull complete
77fa2eb06317: Pull complete
192e2451f875: Pull complete
de57a609c9d5: Pull complete
Digest: sha256:553f64aecdc31b5bf944521731cd70e35da4faed96b2b7548a3d8e2598c52a42
Status: Downloaded newer image for nginx:latest
docker.io/library/nginx:latest
docker@minikube:~$ ^C
docker@minikube:~$ ^C
docker@minikube:~$ minikube exist ssh
-bash: minikube: command not found
docker@minikube:~$ exit
logout
ssh: Process exited with status 127
awais@awais-VirtualBox:~/Documents/devops/assignment3/task5$ kubectl get pods
NAME                                READY    STATUS    RESTARTS   AGE
demo-app-5785849cf5-lvfcb           1/1      Running   0           21m
hello-6f4fd7f4cf-c88xw              1/1      Running   1 (5h10m ago)  41h
hello-6f4fd7f4cf-gjk68              1/1      Running   1           41h
mysql-6974fc57d-cv6l6               1/1      Running   0           5h7m
phpmyadmin-75bf46b7d5-b6f75         1/1      Running   0           5h7m
awais@awais-VirtualBox:~/Documents/devops/assignment3/task5$ kubectl exec -it demo-app-5785849cf5-lvfcb -- sh
# env | grep APP
env | grep DB
APPLICATION_ENV=production
APPLICATION_NAME=MyK8sApp
# DB_PASSWORD=password
DB_USER=root
#
```

TASK#6

1. **Starts the Minikube Kubernetes cluster.**
minikube start
2. **Enables the NGINX Ingress Controller addon for external access.**
minikube addons enable ingress --force
3. **Creates the Persistent Volume (PV) for MySQL storage.**
kubectl apply -f mysql-pv.yaml
4. **Creates the Persistent Volume Claim (PVC) for MySQL.**
kubectl apply -f mysql-pvc.yaml
5. **Deploys the MySQL database using Deployment YAML.**
kubectl apply -f mysql-deployment.yaml
6. **Deploys the PhpMyAdmin application using Deployment YAML.**
kubectl apply -f phpmyadmin-deployment.yaml
7. **Lists all pods to verify MySQL and PhpMyAdmin are running.**
kubectl get pods
8. **Lists all services to verify MySQL and PhpMyAdmin services.**
kubectl get svc
9. **Checks the status of Persistent Volume Claims.**
kubectl get pvc

10. Displays MySQL deployment logs to verify proper startup.
kubectl logs deployment/mysql
11. Opens the hosts file to add a custom domain for PhpMyAdmin.
sudo nano /etc/hosts
12. Adds a mapping of Minikube IP to phpMyAdmin domain.
192.168.49.2 phpmyadmin.local
13. Opens PhpMyAdmin in the browser using the custom domain.
http://phpmyadmin.local

```
awsais@awsais-VirtualBox: ~/Documents/devops/assignment3/task6
Optional:      false
DownwardAPI:  true
QoS Class:     BestEffort
Node-Selectors: <none>
Tolerations:   node.kubernetes.io/not-ready:NoExecute op=Exists for 300s
               node.kubernetes.io/unreachable:NoExecute op=Exists for 300s
Events:
  Type    Reason      Age   From          Message
  ----    -
Normal    Scheduled   18m   default-scheduler Successfully assigned default/phpmyadmin-5b6db79df5-bwjld to minikube
Normal    Pulling     18m   kubelet       Pulling image "phpmyadmin/phpmyadmin"
awsais@awsais-VirtualBox:~/Documents/devops/assignment3/task6$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
mysql-7bc64b954c-4bsx9             1/1     Running   0           3m55s
phpmyadmin-5b6db79df5-bwjld         0/1     ContainerCreating 0           18m
awsais@awsais-VirtualBox:~/Documents/devops/assignment3/task6$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
mysql-7bc64b954c-4bsx9             1/1     Running   0           5m46s
phpmyadmin-5b6db79df5-bwjld         1/1     Running   0           20m
awsais@awsais-VirtualBox:~/Documents/devops/assignment3/task6$ kubectl apply -f ingress-phpmyadmin.yaml
error: the path "ingress-phpmyadmin.yaml" does not exist
awsais@awsais-VirtualBox:~/Documents/devops/assignment3/task6$ kubectl apply -f ingress-phpmyadmin.yaml
error: the path "ingress-phpmyadmin.yaml" does not exist
awsais@awsais-VirtualBox:~/Documents/devops/assignment3/task6$ kubectl apply -f ingress-phpmyadmin.yaml
ingress.networking.k8s.io/phpmyadmin-ingress created
awsais@awsais-VirtualBox:~/Documents/devops/assignment3/task6$ sudo nano /etc/hosts
[sudo] password for awsais:
awsais@awsais-VirtualBox:~/Documents/devops/assignment3/task6$ minikube tunnel
Status:
  machine: minikube
  pid: 21883
  route: 10.96.0.0/12 -> 192.168.49.2
  minikube: Running
  services: []
  errors:
    minikube: no errors
    router: no errors
    loadbalancer emulator: no errors
^Cawsais@awsais-VirtualBox:~/Documents/devops/assignment3/task6$ kubectl get ingress
NAME                                CLASS  HOSTS                ADDRESS      PORTS   AGE
phpmyadmin-ingress                 nginx  phpmyadmin.local    192.168.49.2  80      2m53s
awsais@awsais-VirtualBox:~/Documents/devops/assignment3/task6$
```



```
aws@aws-VirtualBox: ~/Documents/devops/assignment3/task6
aws@aws-VirtualBox:~/Documents/devops/assignment3/task6$ minikube addons enable ingress
⚠ ingress is an addon maintained by Kubernetes. For any concerns contact minikube on GitHub.
You can view the list of minikube maintainers at: https://github.com/kubernetes/minikube/blob/master/OWNERS
  • Using image registry.k8s.io/ingress-nginx/controller:v1.13.2
  • Using image registry.k8s.io/ingress-nginx/kube-webhook-certgen:v1.6.2
  • Using image registry.k8s.io/ingress-nginx/kube-webhook-certgen:v1.6.2
  Verifying ingress addon...
^C
aws@aws-VirtualBox:~/Documents/devops/assignment3/task6$ minikube addons enable ingress --force
⚠ ingress is an addon maintained by Kubernetes. For any concerns contact minikube on GitHub.
You can view the list of minikube maintainers at: https://github.com/kubernetes/minikube/blob/master/OWNERS
  • Using image registry.k8s.io/ingress-nginx/controller:v1.13.2
  • Using image registry.k8s.io/ingress-nginx/kube-webhook-certgen:v1.6.2
  • Using image registry.k8s.io/ingress-nginx/kube-webhook-certgen:v1.6.2
  Verifying ingress addon...
★ The 'ingress' addon is enabled
aws@aws-VirtualBox:~/Documents/devops/assignment3/task6$ kubectl get pods -n ingress-nginx
NAME                                READY   STATUS    RESTARTS   AGE
ingress-nginx-admission-create-pjfv 0/1     Completed 0           9m23s
ingress-nginx-admission-patch-mz2rd 0/1     Completed 1           9m22s
ingress-nginx-controller-9cc49f96f-jr728 1/1     Running   0           9m23s
aws@aws-VirtualBox:~/Documents/devops/assignment3/task6$ kubectl apply -f mysql-pv.yaml
persistentvolume/mysql-pv created
persistentvolumeclaim/mysql-pvc created
aws@aws-VirtualBox:~/Documents/devops/assignment3/task6$ kubectl apply -f mysql-deployment.yaml
deployment.apps/mysql created
service/mysql created
aws@aws-VirtualBox:~/Documents/devops/assignment3/task6$ kubectl apply -f phpmyadmin-deployment.yaml
deployment.apps/phpmyadmin created
service/phpmyadmin created
aws@aws-VirtualBox:~/Documents/devops/assignment3/task6$ kubectl apply -f mysql-service.yaml
kubectl apply -f phpmyadmin-service.yaml
error: the path "mysql-service.yaml" does not exist
error: the path "phpmyadmin-service.yaml" does not exist
aws@aws-VirtualBox:~/Documents/devops/assignment3/task6$ kubectl apply -f mysql-pvc.yaml
error: the path "mysql-pv-pvc.yaml" does not exist
aws@aws-VirtualBox:~/Documents/devops/assignment3/task6$ kubectl apply -f mysql-pv.yaml
persistentvolume/mysql-pv unchanged
aws@aws-VirtualBox:~/Documents/devops/assignment3/task6$ kubectl apply -f mysql-pv.yaml
persistentvolume/mysql-pv unchanged
aws@aws-VirtualBox:~/Documents/devops/assignment3/task6$ kubectl apply -f mysql-pvc.yaml
persistentvolumeclaim/mysql-pvc unchanged
aws@aws-VirtualBox:~/Documents/devops/assignment3/task6$ kubectl apply -f mysql-deployment.yaml
```